Experiment 11: Tic Tac Toe Game

Aim

To implement a two-player Tic Tac Toe game in Python and determine the winner or draw based on the game

state.

Algorithm

1. Start

2. Create a 3 × 3 board.

3. Assign X to the first player and O to the second player.

4. Display the board.

5. Accept a valid position from the current player.

6. Place the player's mark in the selected position.

7. Check rows, columns and diagonals for a winning combination.

8. If a player wins, display the winner and stop.

9. If all cells are filled, declare a draw.

10. Otherwise switch the player and repeat.

11. Stop.

Flowchart

Start

↓

Initialize

↓

Process / Search

↓

Goal / Condition Reached?

n n

Yes No

↓ ↓

Print Result Repeat

↓

Stop

Objective

• Understand game-state representation.

• Implement turn-based decision making.

• Check winning conditions and draw conditions.

Program

```
def print_board(board):
print("\n")
for i in range(0, 9, 3):
print(" | ".join(board[i:i+3]))
if i < 6:
print("--+---+--")
def check_winner(board, player):
wins = [
(0,1,2), (3,4,5), (6,7,8),
(0,3,6), (1,4,7), (2,5,8),
(0,4,8), (2,4,6)
]
return any(all(board[i] == player for i in line) for line in wins)
def tic_tac_toe():
board = [str(i) for i in range(1, 10)]
player = "X"
for turn in range(9):
print_board(board)
move = int(input(f"Player {player}, enter position (1-9): ")) - 1
if move < 0 or move > 8 or board[move] in ("X", "O"):
print("Invalid move. Try again.")
continue
board[move] = player
if check_winner(board, player):
print_board(board)
```

```
print("Player", player, "wins!")
return
player = "O" if player == "X" else "X"
print_board(board)
print("Game Draw!")
# Uncomment to play:
# tic_tac_toe()
```

Output

Sample Output:

Player X, enter position (1-9): 1

Player O, enter position (1-9): 5

...

Player X wins!

Result

The program was executed successfully and produced the expected result.

Conclusion

The experiment successfully demonstrated the corresponding Artificial Intelligence technique and its application